

# AMALKRISHNA PREMKUMAR

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## EXPERIENCE

### Software Engineer

August 2023 – January 2026

Aumsat Technologies LLP (Contract)

Mumbai, India

Sole software engineer owning the full product lifecycle for a suite of GIS applications. Delivered a Dart/Flutter ecosystem of 11 packages (4 Flutter apps + shared libraries), 68K+ LOC and 3,600+ commits.

- Architected and deployed Jal Sanket, a spatial water survey app used by 80+ field workers to log 62K+ geotagged records across 79K+ interactions, shipping the MVP in 10 days under a critical deadline and continuously delivering 14+ releases over 25 days.
- Reduced new app bootstrap time from weeks to hours and to near-zero boilerplate by centralizing 58% of the platform (39K+ LOC across 7 packages) into a shared, modular monorepo core using Dart workspaces.
- Built a four-stage lifecycle map editing engine powering interactive, multi-vertex spatial geometry manipulation with undo/redo, scoped histories and live previews across the platform, using State Machine, Command, and Visitor patterns.
- Authored complex SQL (queries, views, stored procedures) for spatial persistence (PostgreSQL + PostGIS), aggregation and export; executed zero downtime live schema migrations using the expand-contract pattern.
- Engineered PipePulse, a leak management app with a mapping engine that auto-discovers zone hierarchy and renders 10.5K+ map features smoothly via lazy loading and LOD.
- Recovered 4,800+ valid spatial features (96% of the dataset) by implementing a custom RFC 7946 compliant GeoJSON parser and a CLI tool to segregate and isolate corrupted and non-compliant data eliminating silent map rendering failures.

### Software Developer: Institutional Systems

August 2020 – November 2021

KSD's Model College (Dept. of IT & CS)

Dombivli, India

- Independently designed, built and deployed a full-stack academic assessment platform (C#, ASP.NET Core MVC, Blazor Server, PostgreSQL), an initiative commissioned by institution leadership to support remote examinations during COVID-19.
- Architected a hybrid MVC + Blazor Server system with dual-ORM persistence (EF Core + Dapper for high-throughput writes) delivering real-time exam administration.
- Optimized grading workflows by engineering a polymorphic evaluation engine with cross-set fairness validation, cohort result analytics, and spreadsheet-based bulk ingestion/export pipelines.
- Diagnosed and resolved real-time sync issues by iteratively deploying across on-prem IIS and cloud platforms (Digital Ocean App Platform with Docker).
- Automated personalized-certificate generation and distribution during college events for 470+ participants using Google Apps Script (Sheets, Slides, Drive APIs) replacing a fully manual process.

## TECHNICAL SKILLS

**Languages:** C#, Dart, Java, Python, SQL, Go, C/C++, JavaScript/HTML/CSS

**Frameworks:** ASP.NET Core, Blazor, Flutter, Spring Boot, JavaFX

**Databases & Persistence:** PostgreSQL, MySQL, Firebase Firestore, Supabase, PostGIS, ORMs (EF Core, Hibernate)

**Others:** Git, Linux, Docker, Regex, PowerShell, CI/CD (GitHub Actions, Azure DevOps), Jira

## OPEN-SOURCE, INDUSTRY IMPACT & SELECTED PROJECTS

- **Flutter Framework (Google) Contribution:** Landed PR #154073, adding 3 new accessor methods to `ColorScheme` and `TextTheme` streamlining theme access. Engaged with maintainers through the full review process. Credited in the official Flutter 3.27 Release Notes and selected for Google's *Flutter Notable Commits 2024* list.
- **Stack Overflow (Top 0.2% Globally):** Earned **5,000+ reputation** across 230+ technical answers in C#, Python, Java, C++, and more reaching **~984k people**; received community bounties for advanced dynamic programming and debugging solutions.
- **Technical Publications (GeeksforGeeks):** Published authoritative technical deep-dives on modern C# language features (*Pattern Matching*, *Expression-Bodied Members*), consistently ranking **#1–2 in search results** with **40,000+ views**.
- **C# Language Design:** Proposed 3 language features (csharplang #7325, #7028, #9528) with detailed edge-case analysis, drawing engagement from a principal engineer on the C# language design team.
- **NotebooksFX:** Full-stack notebook-style document manager with tree-based navigation & live-preview markdown-based editor: JavaFX desktop client + Spring Boot REST API. Developed a runtime dependency injection container using the Java Reflection API with circular dependency detection, post-construct config, singleton/transient scopes inspired by .NET.
- **Technical Mentorship:** Mentored 15+ undergraduate engineering students 1-on-1 in C/C++ and Python fundamentals & compiled custom interactive Jupyter Notebooks (.ipynb) (2020–2022).

## EDUCATION

**Veermata Jijabai Technological Institute**

Mumbai, IN

Master of Computer Applications

2021 – 2023

**KSD's Model College**

Dombivli, IN

Bachelor of Science in Information Technology

2018 – 2021